

Temporal potential structural connectivity, unprotected
pinch points, and monitoring priorities for free-ranging
cheetahs (*Acinonyx jubatus*) in southern Africa

Liam Crettol

Department of Zoology, Weber State University

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Abstract

[TODO: PLAIN-LANGUAGE GUIDE: briefly explain that cheetah range has contracted and that many cheetahs live outside protected areas. Cite Weise et al. for the range and Marker et al. (2008) for the farmland point.]

The southern African Cheetah (*Acinonyx jubatus jubatus*), has had their range decreased drastically in the past decades. I used a multivariate analysis in which the core variables were fixed at the historical baseline, so the temporal signal came from landscape change. This has only been exacerbated in the past decades due to both natural and anthropogenic reasons, leaving their current extant range majority outside protected areas. This leaves "barriers" like ranchers, roads and other human development in areas where cheetahs will be roaming.

[TODO: PLAIN-LANGUAGE GUIDE: explain that earlier work has mapped suitable habitat and connections, but this study asks how the same modeled network changed through time. The 2026 KAZA multi-species study (Sousa et al., 2026) did not include cheetahs. Do not claim that this is the first cheetah connectivity model; earlier work modeled Asiatic cheetah connectivity.]

Models looking at habitat suitability have been done in the past, but very little predictive modeling of the movement network permeability. One 2026 KAZA (Kavango-Zambezi Transfrontier Conservation Area) multi-species connectivity study excluded the cheetah outright.

[TODO: PLAIN-LANGUAGE GUIDE FOR WHAT WAS DONE: the historical core areas stayed fixed, and the surrounding landscape was modeled for 2012, 2016, 2020, and 2024. Do not give a scenario count until the implemented scenarios have been reconciled with the original protocol.] [TODO: PLAIN-LANGUAGE GUIDE FOR THE MAIN RESULT: modeled connection cost changed only slightly overall. Of 253 core pairs, 161 became somewhat easier and 92 became harder; the middle change was about -1.2 percent. Do not call this an overall connectivity loss.] [TODO: PLAIN-LANGUAGE GUIDE FOR PROTECTION:

outside the cores, about 23 percent of the broader high-current landscape was protected, compared with 31.26 percent of the background area. Protection increased among only the very highest current values.] [TODO: PLAIN-LANGUAGE GUIDE FOR PRIORITIES: 32 of 45 selected links remained in the main priority group, while 13 were kept as places where more field data would be useful. Do not claim that individual monitoring cells were ranked because that index was not built.] [TODO: PLAIN-LANGUAGE CLAIM LIMIT: these results show modeled landscape connections, not confirmed cheetah movement.]

Keywords: landscape connectivity; circuit theory; *Acinonyx jubatus*; protected area coverage; monitoring prioritisation; southern Africa

1 Introduction

The southern African cheetah has had their range substantially reduced, with their current range estimated to be on majority unprotected land. Cheetah range on protected land is governed by centralized park-management initiatives (Buk et al., 2018). Unprotected land poses a different type of challenge, with landowners, local communities and governmental entities involved. This paper looks at landscape connectivity change between 2012 and 2024. The core range of cheetahs was fixed at the 2010-2016 baseline, so modeled changes in permeability reflected landscape changes rather than changes in the core range itself.

Previous research in the region has predominantly focused on habitat suitability, a topic that is already comprehensively addressed in the existing literature. The objective of my study was not to develop an additional suitability model, but rather to identify and evaluate candidate spatial “pinch points” (i.e., constricted linkages) among core areas. The study most comparable to the present work is a recent multi-species connectivity assessment; however, that analysis explicitly excluded cheetah due to insufficient data. Data limitations were therefore explicitly considered and mitigated to the extent possible in this study, and they provide both the primary justification for the existence of this paper and its principal

constraint.

Two primary connectivity approaches are commonly used to model these kinds of frameworks: least-cost connectivity and circuit-theoretic connectivity (Adriaensen et al., 2003; McRae et al., 2008). The least-cost approach links core areas by selecting the route with the lowest cumulative “cost,” meaning the path of least resistance between cores, based on weight values assigned to each multivariate resistor. In contrast, the circuit-theoretic approach represents the landscape as a resistor network, sends a “current” through it, and highlights redundancy across the broader landscape, relying on the same multivariate resistors. A key limitation of both methods is that you must first assign a resistance “score” or “weight” to every cell (1 km in my case). Zeller et al. (2012) demonstrated that even reasonable alternative assumptions about resistance values can influence results more strongly than the choice of connectivity method itself. For that reason, rather than treating one resistance surface as definitive, I varied the weights along two axes. The first changed the balance among the anthropogenic pressures, comparing my primary weighting against an equal weighting and an infrastructure-emphasis weighting. The second changed how much the surface leaned on vegetation relative to anthropogenic pressure and terrain, at 10, 20 and 40 percent. The vegetation-balanced surface combined with the documented fence treatment carries the primary results, and the remaining combinations are reported as sensitivity rather than as alternative answers. A link is only described as robust here if it survives them. This helps to minimize bias from the selection of resistor weights.

[TODO: PLAIN-LANGUAGE GUIDE: add one sentence explaining that the completed scenarios do not match the original RES-01 to RES-06 labels exactly. Add Zeller et al. (2018) beside Zeller et al. (2012) for alternative weighting and Beier et al. (2008) for how modeling choices can affect corridor results.]

[TODO: Write one overall aim identifying the region, fixed historical cores, and the 2012–2024 comparison. Frame the objective around measuring change without assuming its direction.]

[TODO: Give temporal change, link robustness, and protected-area coverage their own objective. Map each objective to a completed analysis and a Results subsection. Decide whether monitoring implications belong under link screening or in Discussion rather than promising a separate monitoring index.]

[TODO: For each objective, decide whether a hypothesis or a descriptive research question fits the work. Distinguish what you assessed, the relationship you proposed, and the measurable pattern you expected.]

[TODO: Identify the main connectivity measure and explain which direction means greater difficulty of connection. Keep effective resistance, least-cost path cost, and mapped current distinct. Put calculation details in Methods.]

[TODO: Separate H1 into two assessable predictions: whether loss is concentrated among links, and whether loss is associated with link importance. Define concentration, minority, and high betweenness if retaining those terms. Do not choose cutoffs to match the results.]

[TODO: Specify the population for each comparison: all 253 core pairs or the 45 selected network links. Avoid testing a network-wide prediction only on the screened priority subset.]

[TODO: Check the meaning of baseline betweenness against the implemented unweighted graph. Explain what the importance measure represents and keep network position separate from measured temporal change.]

[TODO: Locate the evidence evaluating each part of H1. State what would count against each prediction. Distinguish a descriptive assessment from a formal test, and an unsupported prediction from one that was not tested. Avoid a hypothesis that simply repeats how pressure inputs were converted to resistance.]

[TODO: Define high-value connectivity area using a specific modeled quantity and comparison domain. Identify which cells enter the outside-core analysis and which current thresholds are being compared.]

[TODO: Specify the background against which disproportionate outside coverage is assessed. An outside share greater than half is not sufficient by itself; identify the matching

area and denominator.]

[TODO: Align H2 with the protected-area analysis actually reported. Decide whether to retain a hypothesis with an explicit assessment criterion or use a descriptive coverage question. Do not promise a significance test that the Methods no longer uses.]

[TODO: Explain how evidence that changes across current thresholds will be handled. Keep the original expectation separate from patterns noticed during exploration; do not select a threshold because it supports the prediction.]

[TODO: Clarify that the protected-area overlay is a fixed protection context, rather than a reconstruction of protection at each historical date. Leave data-release and filtering details for Methods.]

[TODO: Define importance separately from robustness. Distinguish network importance, stability of change direction, and stability of route location. Decide which of these H3 actually predicts.]

[TODO: Limit the scope to the resistance assumptions and sensitivity checks actually completed. Describe the implemented link-level screening in Methods; do not imply that a cell-level agreement map, alternative core definitions, or multiple grid resolutions were tested.]

[TODO: Define limited subset if keeping it as a prediction. Do not adjust its meaning to fit the resulting count. Verify whether transboundary links were explicitly classified and assessed before retaining that term in H3.]

[TODO: State what evidence would fail to support H3. Separate failure of the predicted pattern from the usefulness of the sensitivity analysis; uniformly high agreement does not make the comparison useless.]

[TODO: Keep numerical results and support judgments in Results and Discussion. Describe questions developed after examining results as exploratory.]

[TODO: Check that every objective has a matching analysis and that every retained prediction has a clear measure, comparison, and assessment criterion. Keep procedural

details in Methods and retain the terminology boundary below.]

Throughout, potential structural connectivity refers to modelled landscape permeability between fixed range cores. It is not evidence of cheetah movement, dispersal, or gene flow. Where a location is described as a pinch point, this means a location of concentrated modelled current or a link-level bottleneck, not an observed crossing.

2 Methods

2.1 Study area and analysis extent

The study area was defined as the southern African cheetah range (Weise et al., 2017) plus a buffered analysis extent. All layers were aligned to a fixed equal-area projection and a 1 km snap grid, so every raster was directly comparable cell for cell. Cores were derived from the historical confirmed-range baseline and held constant across 2012, 2016, 2020, and 2024. The final network consisted of 23 cores and 45 selected neighboring core pairs.

2.2 Rationale for the working resolution

I used a 1-km working resolution because it represented a compromise between the rougher 10-km occurrence baseline and the finer environmental covariates, which ranged from 30 m to 1 km. This resolution was fine enough to identify meaningful points in the landscape while avoiding the implication of a level of accuracy that the occurrence evidence alone could not provide.

2.3 Range cores

I defined a core area as a contiguous patch of cells that were confirmed presence, with a density of ≥ 0.51 cheetahs per 100 km^2 . The final network consisted of 23 cores, which were held constant across all four temporal snapshots to avoid inventing core movement in the

absence of comparable range reassessment.

I represented these 23 core areas as fixed nodes across all four temporal snapshots (2012, 2016, 2020 and 2024). I took each core area and calculated the Euclidean distance to the nearest neighbor. This yielded 45 straight-line links. These straight-line links defined the lines used to get the least cost paths based on the added resistance surface rasters.

2.4 Covariates and their temporal treatment

The predicted temporal resistance values came from the continuous rasters (built-up fraction (human development) from GHSL, nighttime lights from VIIRS, and VCF (vegetation) from MODIS). Categorical land cover was used for masking and description only, because annual categorical features can switch classes between years for reasons unrelated to overall landscape change.

I also used the static covariates (roads, livestock, terrain, hydrography, fences, protected areas, and core definitions). The static covariates did not contribute *temporal* signal to the resistance surface; however, they did contribute *spatial* signal to the resistance surface. The static covariates were used to describe the landscape and to provide context for the dynamic covariates.

2.5 Resistance scenarios

Resistance values were assigned to represent the relative difficulty of movement through each landscape feature. The terrain component was derived from slope, with steeper slopes assigned higher resistance values. I compared alternative vegetation weights because vegetation cover can significantly influence cheetah movement, [TODO: add a cite for this] and different weighting schemes allowed for sensitivity analysis of just this factor. the final balanced map [TODO: figure?] had a weight of 60% for human pressure, 20% for vegetation, and 20% for terrain. Within human pressure, built-up land, roads, and livestock each received 30%, while nighttime lights received 10%. Available fence data was incorporated into the resis-

tance model, with fence cells increasing the combined resistance by a multiplier from 1 to 25. The final resistance surface was a weighted sum of the terrain, vegetation, and anthropogenic components. The anthropogenic component was derived from built-up land, roads, livestock density, and nighttime lights. The terrain component was derived from slope, with steeper slopes assigned higher resistance values. The vegetation component was derived from the VCF data, with denser vegetation assigned lower resistance values.

[OUTLINE]

Terrain component. The DEM was not used directly as resistance; slope was derived from it. Mean slope carries the primary terrain surface, maximum slope retained for sensitivity and context. Terrain resistance function: slope $\leq 10^\circ$ low resistance; 10 to 30° linearly increasing; $\geq 30^\circ$ capped high; resistance rescaled to 1 to 10.

Original primary temporal surface. Anthropogenic component 80% of total, terrain 20%. Within the anthropogenic component: built-up 30%, roads 30%, livestock 30%, nighttime lights 10%.

Three internal weighting models compared: primary, equal, infrastructure-emphasis.

Vegetation correction. The first temporal model omitted VCF from the resistance calculation, which is a mismatch with the predeclared protocol. Three further sensitivity scenarios were added, keeping the original anthropogenic-only surfaces intact rather than replacing them:

- vegetation-low: 10% vegetation, 70% anthropogenic, 20% terrain
- vegetation-balanced: 20% vegetation, 60% anthropogenic, 20% terrain
- vegetation-high: 40% vegetation, 40% anthropogenic, 20% terrain

The vegetation proxy uses continuous VCF fields to represent sparse and bare cover. It does not claim to measure fine-scale habitat edge or observed movement preference, and should be described that way.

Reconciliation still needed: these six weighting models are not the same object as the predeclared RES-01 to RES-06 scenario set. Either map each one onto a predeclared scenario ID or state plainly in the text that the implemented set differs and why.

Resistance values were assigned to represent the relative difficulty of movement through each landscape feature. The terrain component was derived from slope, with steeper slopes assigned higher resistance values. This had to be done carefully however, given cheetahs do have an affinity for some sloped terrains [TODO: PLAIN-LANGUAGE COMPARISON: Moqanaki and Cushman (2017) treated slopes near 15 percent as easiest for movement and assigned more resistance above or below that point. Explain that this study instead used chosen slope ranges as a sensitivity assumption because no local GPS data were available to estimate the exact response.]. I compared alternative vegetation weights because vegetation cover can significantly influence cheetah movement, and different weighting schemes allowed for sensitivity analysis of this factor.

2.6 Connectivity modelling

[TODO: PLAIN-LANGUAGE WRITING GUIDE. Least-cost paths: for each starting core, ArcGIS calculated the accumulated cost of moving through every reachable cell. A path was then drawn from each connected core back to the starting core by following the lowest accumulated cost. Reusing one cost calculation for several destinations gives the same pairwise paths with fewer repeated calculations. Circuitscape: Circuitscape 5.17.1 (Anantharaman et al., 2020) treated the landscape like an electrical network and spread current across all possible routes between each pair of cores. All 253 core pairs were solved in each of four years, for 1,012 pair comparisons, with no failures. Why cores were removed from reported current: cores are where modeled current enters and leaves, so they automatically receive the highest values. Removing the core cells after calculation lets the maps focus on land between cores. This improves what is reported, but the cores still influenced the calculation. Why a core-to-core method was used: the study began with defined core areas, so the question was how those known areas connect. An all-direction method such as Omniscape (Landau et al., 2021) answers a different question.]

Corridor definition. [TODO: PLAIN-LANGUAGE WRITING GUIDE: do not draw a hard boundary and call everything inside it a confirmed corridor. Report the full continuous current surface. Percentile classes are used only to make the maps easier to read and to compare protected-area coverage. Percentiles are calculated after core cells are removed, and the class boundaries were chosen before looking at the protected-area result.]

2.7 Network structure

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What was measured: edge betweenness counts how often a link lies on the shortest network route between pairs of cores. A high value means the link helps hold many parts of the modeled network together. The value was calculated on the 45-link network and divided by the 253 possible core pairs. How links were chosen: the network combined each core's three nearest neighbors with a minimum spanning tree, which is the smallest set of links needed to keep the core network connected. Why rankings are uncertain: allowing two, three, four, or five nearest neighbors created 31, 45, 61, or 77 links, and the highest-ranked links changed. Link 17–24 was the only bridge under the three-neighbor choice; removing a bridge separates the network into two pieces. What was dropped: PC and dPC (Saura & Pascual-Hortal, 2007) require an assumed relationship between distance and dispersal. No local dispersal telemetry was available, so those more detailed-looking numbers were not calculated. Cite Keeley et al. (2021) when explaining that the metric should match the research question.]

[OUTLINE]

edge betweenness computed on the 23-core network, link importance ranked on normalised edge betweenness. PC and *dPC* not yet computed, and dispersal distance is still the 100 km placeholder.

2.8 Temporal comparison

[TODO: PLAIN-LANGUAGE WRITING GUIDE: compare the same core pairs, model assumptions, and 1-km grid in 2012, 2016, 2020, and 2024. This keeps the comparison con-

sistent so the modeled differences come from the input layers that changed through time. Define absolute change as later value minus earlier value. Define percent change as that difference divided by the earlier value, multiplied by 100.]

2.9 Protected-area coverage

[TODO: PLAIN-LANGUAGE WRITING GUIDE. Question: what share of the modeled connecting landscape outside the cores falls inside protected areas? Data preparation: use the August 2026 WDPA release (UNEP-WCMC & IUCN, 2026). Keep terrestrial and coastal protected-area polygons, remove proposed areas, points, and marine-only areas, and merge overlaps. This left 2,301 records in the study extent. Comparison area: 31.26 percent of the usable outside-core area was protected. Cells with zero current were excluded because they could never enter a high-current group. Why no p-value: the answer changed from significant to not significant when the assumed spatial block size changed. Report the descriptive percentages and note that the same general pattern appeared in all four years and under several core-buffer sizes.]

2.10 Robustness and uncertainty

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What "robust" means here: a link had to pass three checks. First, both alternative weightings had to keep at least 80 percent of the route within 5 km of the main route; 34 of 45 passed. Second, the fence scenarios could not substantially change the link; this removed two more from the group. Third, the direction of change had to stay the same under all vegetation weightings. The two links that failed this last check had already been removed. The final groups were 32 main-priority links and 13 uncertain links. Important distinction: the 80-percent-within-5-km rule is not the cell-by-cell agreement test proposed in the original protocol, even though both use the number 80 percent. What was not completed: alternative core definitions, 2-km and 5-km grids, and a cell-by-cell agreement map. Separate location uncertainty from cost uncertainty: 49 of

180 route-and-year comparisons had less than 80 percent overlap within 1 km, including 21 main-priority links. Link 17–24 had no overlap in 2020. This does not mean the connection was unimportant; it means the exact mapped route was uncertain.]

[OUTLINE]

- 90 weight comparisons evaluated.
- 34 of 45 core pairs robust under the tested alternative weights, 11 sensitive.
- Fence scenarios changed 5 links: 1–13, 1–17, 9–13, 13–17, 27–30.
- 32 links classified primary-priority eligible; 13 retained as uncertainty and data-collection priorities.

Two things to resolve before writing: state the numeric robustness rule actually applied and whether it matches the predeclared $k/n = 0.8$; and reconcile the 34 robust pairs with the 32 primary-priority-eligible links, since those are different counts and a reader will notice.

2.11 Independent plausibility assessment

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What was planned: compare the model with separate Limpopo tracking data and Namibia camera-trap data. What was actually done: compare early and late occurrence records from the same archive used to create the cores. Because the data are not independent, call this a limited plausibility check rather than validation. Numbers available: after thinning to one record per 10-km cell, 2010–2012 had 413 cells and 2013–2016 had 450. Their overlap score was 0.239. There are no records after 2016, so this check says nothing about 2020 or 2024. Decision still needed: either complete the planned outside-data check or clearly state that it was not completed.]

2.12 Monitoring prioritisation

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What was planned: combine six measurements into one score for every 10-km cell and test three sets of weights. What was built

instead: a link-level table that keeps each kind of evidence separate. Its columns describe how important the link is to the network, how its modeled cost changed, how stable its mapped route was, whether fences affected it, and how much of it is protected. Why this is more honest: combining everything into one score would require deciding, for example, how much route instability is equal to a change in protection. No evidence-based exchange rate was available. State clearly that no combined monitoring score was calculated.]

[OUTLINE]

link ranking currently rests on normalised edge betweenness plus the robust / sensitive split, with 32 primary-priority-eligible and 13 uncertainty-priority links. The full MCDA with three weight sets is not yet built.

2.13 Software and reproducibility

[TODO: PLAIN-LANGUAGE WRITING GUIDE: list the software without explaining every technical setting in the main text. Verified versions are ArcGIS Pro 3.7.2 (build 1901), ArcGIS Python 3.13.13, Julia 1.12.7, and Circuitscape.jl 5.17.1; see Supplementary Table 5. Google Earth Engine supplied and aligned the satellite data. Most ArcGIS processing used Python scripts rather than ModelBuilder. Add the final archive link and state which source datasets cannot legally be redistributed.]

[OUTLINE]

ArcGIS Pro Spatial Analyst via arcpy (`CostDistance`, `CostPathAsPolyline`); Google Earth Engine for SRTM, GHSL, VIIRS, MODIS MCD12Q1 and VCF acquisition and harmonisation. Record versions.

3 Results

3.1 Connectivity change, 2012–2024

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What happened: I compared all 253 possible pairs of cores. Overall, the modeled cost of moving between cores changed only a little from 2012 to 2024. The middle change was -1.224 percent. A negative value means the model became slightly easier to move through; it does not mean that cheetahs actually moved more. Of the 253 pairs, 92 became harder and 161 became easier. The largest increase was +13.1 percent for the link between cores 33 and 38. What the map adds: outside the cores, the places with the highest modeled current were mostly in the same locations through time. Their overlap scores were 0.91 to 0.93, where 1 would mean perfect overlap. What not to claim: do not call this an overall connectivity loss, observed movement, or a country-by-country result. The analysis did not assign each core pair to a country. For H1, report that 18 of the 45 selected links became harder and 27 became easier, but that change was not meaningfully related to baseline betweenness ($\rho \approx 0.01$). Therefore H1 was mixed rather than supported. Use Figure 3.]

3.2 Priority links and sensitivity checks

[TODO: PLAIN-LANGUAGE WRITING GUIDE. What happened: 32 of the 45 selected links passed the tested sensitivity screens and remained in the main priority group. Network importance was ranked separately using betweenness. The other 13 links were kept as uncertain links where more field information would be useful. The weighting test kept 34 links, and the fence test removed two more. The vegetation test did not remove any additional links from the remaining group. What it means: the main group is not a small handful of links; it contains most of the network. That only partly matches the original expectation that priorities would be limited to a small subset. What not to claim: these links are modeled connections between cores, not confirmed animal travel routes. Use Table 2.]

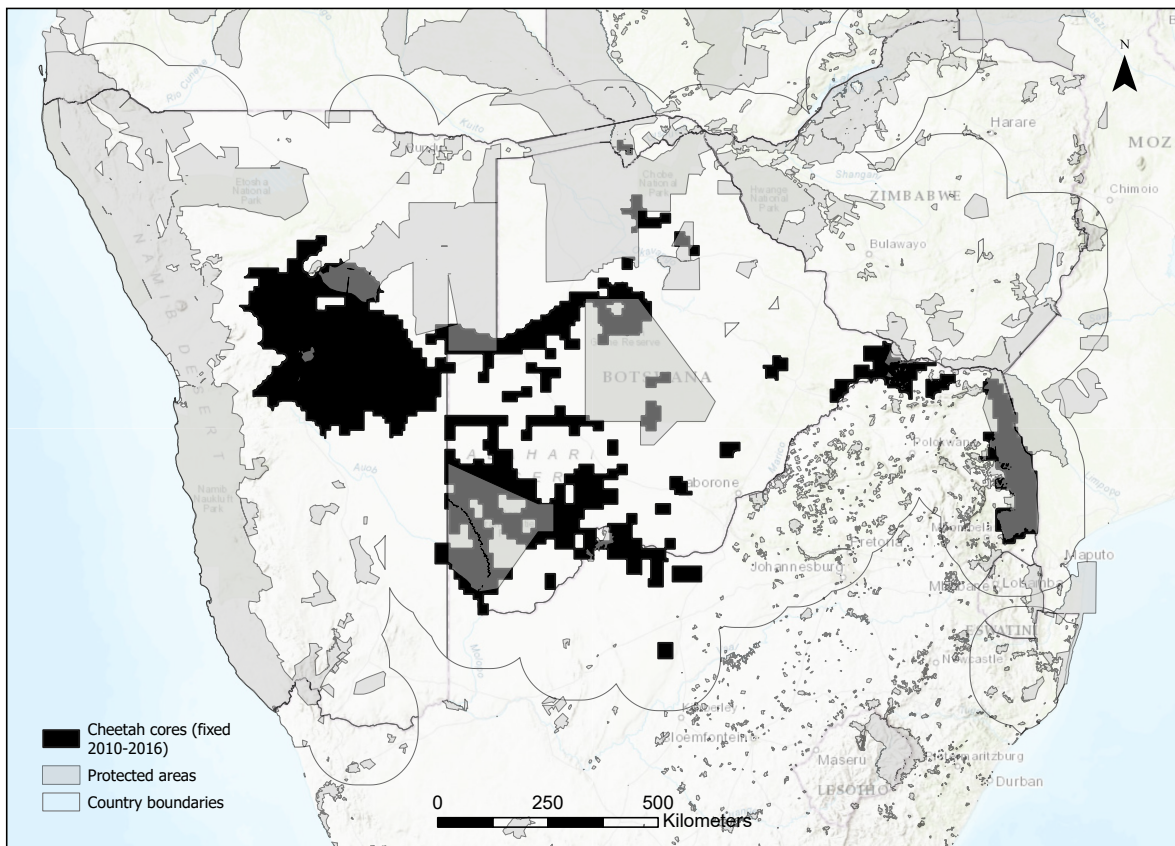


Figure 1: Study extent, fixed historical range cores, and core groups used to define the modeled connectivity network. Core groups are an analysis construct and do not imply separate populations.

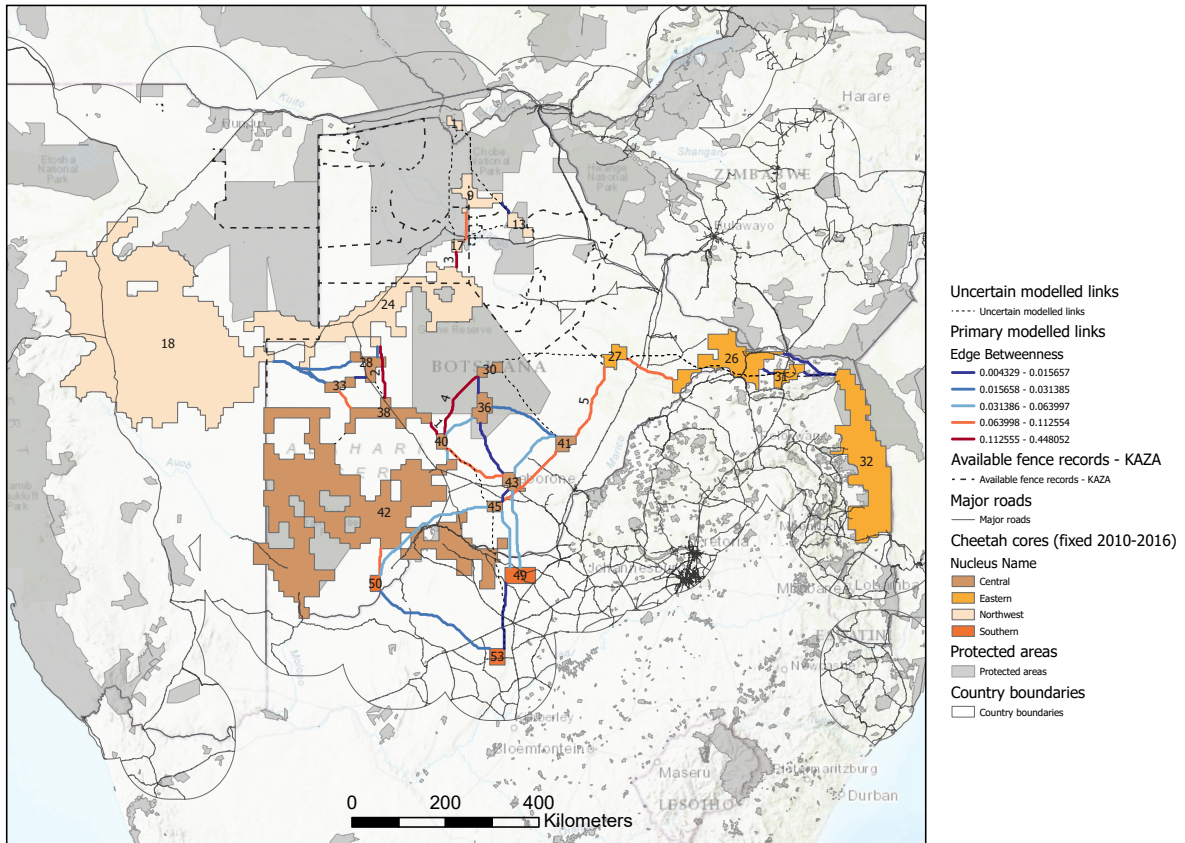


Figure 2: Modeled core-to-core links with transportation and available fence context. Lines represent network links used to define least-cost analyses; they are not observed animal paths.

[TODO: PLAIN-LANGUAGE WRITING GUIDE FOR THE SENSITIVITY TESTS. Main lesson: the total modeled cost of a connection was usually fairly stable, but the exact line taken by the least-cost path was less stable. Of 180 route-and-year comparisons, 49 had less than 80 percent overlap within 1 km. This affected 21 of the 32 priority links. The route between cores 17 and 24 had no overlap in 2020. Vegetation assumptions: 43 of 45 links kept the same direction of change under all three vegetation weightings. However, the size of the change grew as vegetation was given more weight: the middle absolute change increased from about 3.6 to 9.3 percent. Network-design assumption: changing how many neighbors each core could connect to changed the number of links and changed which links ranked highest. A bridge is a link whose removal splits the network into separate pieces; link 17–24 was the only bridge when three nearest neighbors were used. What not to claim: there is no completed cell-by-cell agreement map, and no single ranking should be presented as certain. State that H3 was only partly evaluated: a robust group existed, but it contained a majority of the selected links and the full planned sensitivity grid was not completed.]

[TODO: PLAIN-LANGUAGE WRITING GUIDE FOR THE OCCURRENCE CHECK. What happened: after thinning records to one 10-km cell, the early period had 413 occupied cells and the later period had 450. Their overlap score was 0.239, showing that the two sets were quite different. Later records occurred in slightly higher modeled resistance, and a larger share fell inside the fixed cores. Almost no records were close to the priority paths: zero early cells and three later cells. Why this is a weak check: the same occurrence archive helped create the cores, so this is not an independent test. There are also no occurrence records after 2016, so it cannot check the 2020 or 2024 results. Report it honestly as a limited comparison, not as validation.]

[TODO: Protocol reporting only: the planned cell-level monitoring index was not built and H4 was not tested. Explain this once in Appendix A, not as a repeated priority-link result.]

3.3 Protected-area coverage of priority connectivity

[TODO: PLAIN-LANGUAGE WRITING GUIDE. Why the cores were removed: the model forces current into and out of the cores, so the cores automatically receive very high current. Including them would make protection look better without telling us how well the land between cores is protected. What happened outside the cores: 31.26 percent of the comparison area is protected. Only 23.27 percent of the area above the 80th current percentile was protected in 2024. Coverage rose to 25.54 percent above the 90th percentile, 33.45 percent above the 95th, and 44.63 percent above the 99th. In simple terms, the very strongest concentrations often reached protected areas, while the broader connecting landscape around them was less protected. What not to claim: do not report a significance p-value because it changed when the size of the spatial blocks changed. Do not call modeled current observed movement. Use Table 3. State that H2 was not supported under its full rule because the direction changed across current thresholds and the significance result was not stable.]

4 Discussion

4.1 Connectivity change and comparison with previous work

[TODO: PLAIN-LANGUAGE GUIDE: begin with the simplest overall result: the modeled ease of moving between cores changed only slightly overall, but some individual links changed more than others. Immediately state that the model describes landscape structure and does not show where cheetahs actually moved.]

[TODO: PLAIN-LANGUAGE GUIDE: describe where modeled connections became easier or harder and which landscape changes occurred in the same places. Say "occurred in the same places," not "caused." The model found mixed change rather than a broad decline. Do not describe connectivity by country because core pairs were not assigned to individual countries.]

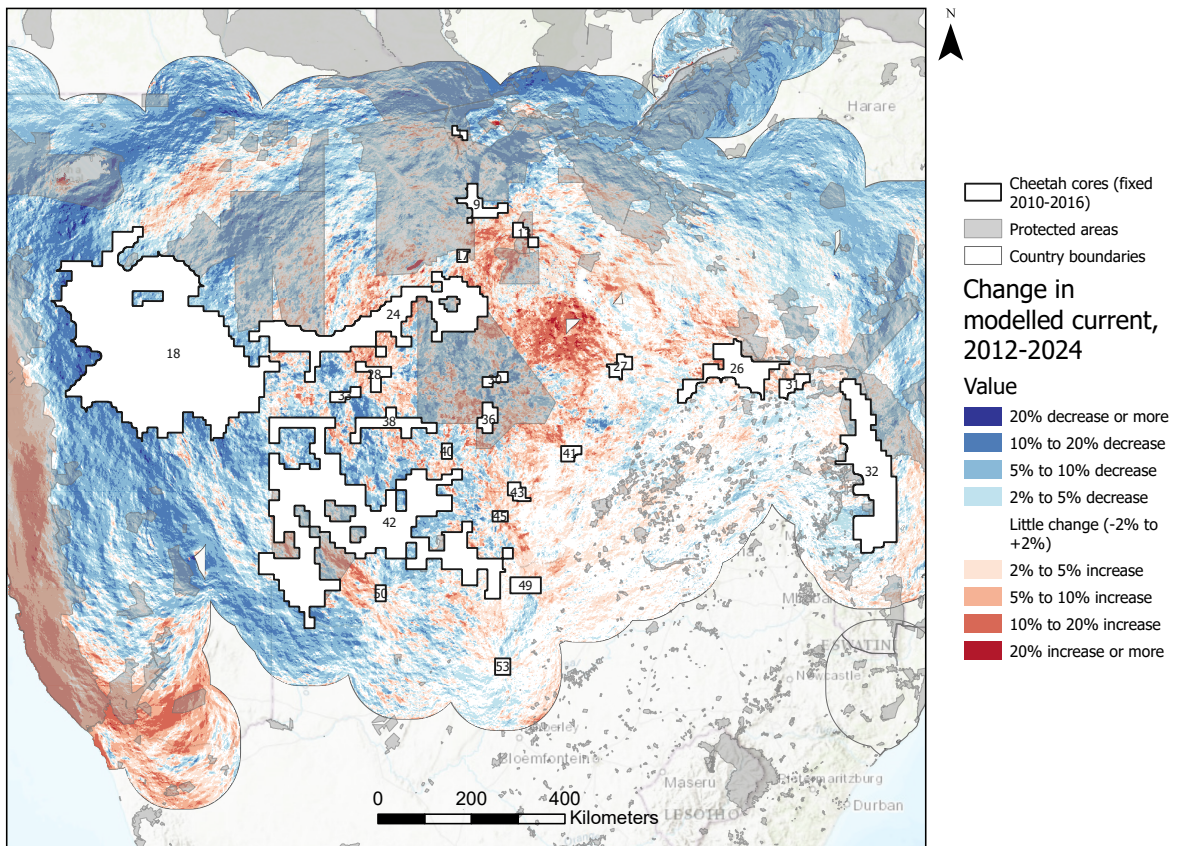


Figure 3: Modeled current change between 2012 and 2024. The map is a descriptive comparison of structural-flow outputs and should not be interpreted as observed movement or a validated corridor.

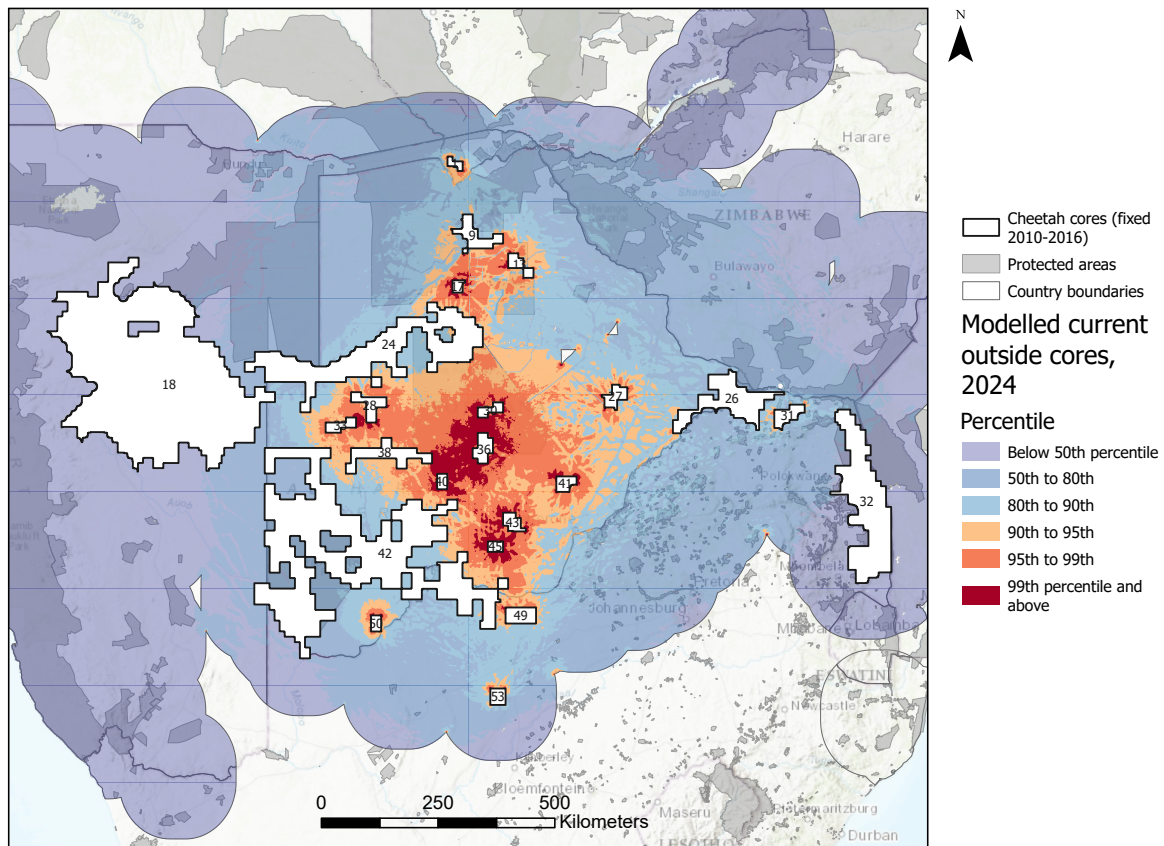


Figure 4: Modeled current concentration for 2024, with the fixed core network shown for orientation. Current concentration is structural-flow evidence, not occurrence probability.

Table 1: Covariates, native resolution, source years, and static/dynamic treatment.

Covariate	Source	Treatment	Temporal role / notes
Built-up fraction	Global Human Settlement Layer (GHSL)	Dynamic	2010, 2015, 2020, 2025 products matched to snapshots
Nighttime radiance	Visible Infrared Imaging Radiometer Suite (VIIRS), VNL V2	Dynamic	2013, 2016, 2020, 2024 products
Tree cover	Moderate Resolution Imaging Spectroradiometer (MODIS), MOD44B	Dynamic	Annual snapshot fields
Non-tree vegetation	Moderate Resolution Imaging Spectroradiometer (MODIS), MOD44B	Dynamic	Annual snapshot fields
Bare cover	Moderate Resolution Imaging Spectroradiometer (MODIS), MOD44B	Dynamic	Annual snapshot fields
Road density	Global Roads Inventory Project, version 4 (GRIP4)	Static	Spatial pressure/context
Livestock density	Gridded Livestock of the World, version 4 (GLW4)	Static	Combined cattle, goats, and sheep exposure
Terrain / slope	Shuttle Radar Topography Mission (SRTM)	Static	Slope-derived resistance
Hydrography	Hydrological data and maps based on Shuttle Elevation Derivatives at multiple Scales (HydroSHEDS)	Static	Water/context mask
Veterinary fences	World Wide Fund for Nature (WWF), Kavango–Zambezi Transfrontier Conservation Area (KAZA); Kruger references	Scenario	Sensitivity/context, not assumed impassable

Note: Product shorthand retained in the source names is defined at first use: VNL = VIIRS nighttime lights; V2 = product version 2; MOD44B = MODIS vegetation-continuous-fields product.

Table 2: Ten highest-ranked modeled priority links shown from the 32-link primary-priority set.

From	To	Rank	Betweenness	Type	Distance (km)	Change (%)
38	40	1	0.409	KNN+MST	120.6	+4.78
24	38	2	0.316	KNN	202.3	+7.44
17	24	3	0.300	KNN+MST	208.0	-5.78
30	40	4	0.238	KNN	169.2	+3.55
27	41	5	0.103	KNN+MST	196.8	-6.45
41	45	6	0.090	KNN	183.6	-5.50
33	38	7	0.079	KNN	108.1	+13.12
9	17	8	0.079	KNN+MST	109.0	-5.26
26	27	9	0.079	KNN+MST	221.8	-5.18
40	43	10	0.077	KNN	159.6	-1.56

Notes: The table shows the ten highest-ranked members of the 32-link primary-priority set; links are modeled structural network relationships, not observed animal routes. KNN = k -nearest-neighbor link; MST = minimum spanning tree link. “KNN+MST” identifies links retained by both network-construction rules.

Table 3: Protected-area coverage of modeled high-current classes.

Year	Current percentile	Cells	Protected (%)	Background (%)	Difference (pp)
2012	80	418,295	24.23	31.26	-7.03
2012	90	209,148	26.31	31.26	-4.95
2012	95	104,574	35.57	31.26	+4.31
2012	99	20,915	44.17	31.26	+12.92
2024	80	418,292	23.27	31.26	-7.99
2024	90	209,146	25.54	31.26	-5.72
2024	95	104,573	33.45	31.26	+2.19
2024	99	20,915	44.63	31.26	+13.37

[TODO: Final note: values use the outside-core, positive-current domain. Percentage-point differences are relative to the matched background. Do not report a permutation p -value because inference was sensitive to the assumed spatial block size.]

Table 4: Overview of completed modeled-connectivity results.

Metric	Value	Interpretation
Final modeled-current snapshots	4 (2012, 2016, 2020, 2024)	253/253 focal-region pairs connected each year
Fixed density cores	23	Historical baseline, fixed across snapshots
Selected neighboring links	45	Core-pair comparison set
Primary-priority links	32	Robust under weight/fence screens
All-pair median effective-resistance change	-1.224%	2012–2024
Selected-link median change	-1.557%	2012–2024
Primary-priority median change	-2.618%	2012–2024
Largest all-pair increase	+13.119%	Link 33–38
Largest all-pair decrease	-8.243%	Link 43–45

[TODO: PLAIN-LANGUAGE GUIDE: compare the general mapping approach with the elephant and multi-species studies (Brennan et al., 2020; Naidoo et al., 2024) and with the fence study (Osipova et al., 2018). Explain that similar mapping tools can be used for different species, but the numerical resistance values cannot simply be copied because each species responds differently to roads, vegetation, fences, and people.]

4.2 Protected areas and conservation context

[TODO: PLAIN-LANGUAGE GUIDE: the strongest modeled concentrations often pass through protected areas, but the wider land connecting those areas is less protected. Explain that conservation therefore also depends on farms, communal lands, and other land outside formal parks. This does not prove that protected areas failed. Also note that parks are often located on land with lower agricultural value, which may partly shape the pattern.]

4.3 Priority links and field-survey needs

[TODO: PLAIN-LANGUAGE GUIDE: links that stayed important under several assumptions are the strongest candidates for conservation attention. Links that changed substantially between assumptions are useful places for field surveys because better local data could change the result. Agreement between scenarios is not complete proof because the scenarios share many of the same input layers.]

[TODO: PLAIN-LANGUAGE GUIDE: there is no completed score that ranks individual survey cells. Instead, explain how an organization could use the 13 uncertain links to choose general survey areas, then adjust the order based on access, cost, safety, and local knowledge. Link 17–24 is especially useful for checking because it is the only link holding two parts of the three-neighbor network together, yet its mapped route was highly unstable and had no overlap in 2020.]

5 Limitations

[TODO: SENTENCE 1: Say that the model shows where movement may be easier or harder, but it does not prove where cheetahs actually traveled. SENTENCE 2: Say that the cores and several landscape layers were kept constant through time, allowing a fair comparison but limiting what could change between years. SENTENCE 3: Say that the resistance values came from published research rather than GPS movement data and that differences in data coverage, rainfall, and source years add uncertainty. SENTENCE 4: Say that the mapped paths are broad planning areas rather than exact travel routes and that the results should be tested before use at another scale or in another region.]

6 Conclusion

[TODO: PLAIN-LANGUAGE GUIDE: use only three or four sentences. First, say that modeled connectivity changed only slightly overall and that individual links changed in both directions. Second, say that the strongest concentrations often reached protected areas but the wider connecting landscape was less protected. Third, state that 32 links were retained as main priorities and 13 were identified as places needing more field information. End by reminding the reader that these are modeled landscape connections, not observed cheetah routes. Do not add new results here.]

Data availability

[TODO: PLAIN-LANGUAGE GUIDE: add the final repository or archive link and DOI. State that the scripts and outputs created by the project are available there. Explain that raw protected-area data must be downloaded from Protected Planet under its license and that cheetah occurrence locations can only be shared at the detail allowed by their source.]

Acknowledgements

[TODO: PLAIN-LANGUAGE GUIDE: list the department, advisor, data providers, and any organization or person who reviewed the project or helped interpret priorities.]

Conflicts of interest

I declare no conflicts of interest.

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A Analysis protocol as predeclared

[TODO: PLAIN-LANGUAGE GUIDE FOR THE PROTOCOL APPENDIX. The saved protocol is tagged `protocol-v1.0` in the `cheetah-connectivity` repository at commit `d7de8675a074dfbffc0c7c856bb5645e1e13867d`, dated 18 August 2026. However, the file itself still says version 0.1 draft. Do not change the old file. Explain that the Git tag marked the saved version even though its internal label was not updated. Then list, in simple terms, what changed between the plan and the final analysis: the core definition, sensitivity tests, definition of a robust link, scenario names, network measurements, protected-area comparison, occurrence check, monitoring index that was not built, and 2030 scenarios that were added later. Include the objective and hypothesis outcomes: O1 was completed; O2 was

narrowed; O3 was replaced rather than completed; H1 was mixed; H2 did not pass its full rule; H3 was only partly tested and did not match the “limited subset” prediction; and H4 was not tested.]

B Supplementary material

[TODO: PLAIN-LANGUAGE GUIDE: the tables below preserve details that are useful for checking the analysis but are too long for the main paper. They cover which source year matched each modeled year, strongly related predictors, overlap of high-current areas, software versions, vegetation assumptions, route stability, and the 2030 scenarios. Do not add tables for different map resolutions or a cell-level monitoring score because those analyses were not completed.]

Table 5: [TODO: Final caption: software environment used for the completed analysis.]

Component	Version	Verification
ArcGIS Pro	3.7.2 (build 1901)	<code>arcpy.GetInstallInfo()</code>
ArcGIS Python	3.13.13	ArcGIS Pro environment
Julia	1.12.7	<code>julia -version</code>
Circuitscape.jl	5.17.1	Julia package environment
Google Earth Engine	Cloud service	Acquisition and harmonisation

Table 6: [TODO: Final caption: temporal matching of dynamic covariates to the four analysis snapshots.]

Covariate	Source	Treatment	2012 snapshot	2016 snapshot	2020 snapshot	2024 snapshot
Built-up fraction	GHSL	Dynamic	2010 (−2)	2015 (−1)	2020 (0)	2025 (+1)
Nighttime radiance	VIIRS VNL V2	Dynamic	2013 (+1)	2016 (0)	2020 (0)	2024 (0)
Tree cover	MODIS MOD44B C6.1	Dynamic	2012 (0)	2016 (0)	2020 (0)	2024 (0)
Non-tree vegetation	MODIS MOD44B C6.1	Dynamic	2012 (0)	2016 (0)	2020 (0)	2024 (0)
Bare ground	MODIS MOD44B C6.1	Dynamic	2012 (0)	2016 (0)	2020 (0)	2024 (0)
Land cover	MODIS MCD12Q1 C6.1	Context only	2012 (0)	2016 (0)	2020 (0)	2024 (0)

[TODO: Final note: values in parentheses are source-year minus nominal-year offsets. Define GHSL, VIIRS, VNL, and MODIS here.]

[TODO: PLAIN-LANGUAGE DATA CHECK: every value in these tables should come from a CSV in `tables/source_data/`. Never copy a number from a screenshot or estimate

Table 7: [TODO: Final caption: predictor pairs with absolute Spearman correlation of at least 0.75.]

Predictor pair	Spearman ρ
Tree cover–bare ground	−0.877
Non-tree vegetation–bare ground	−0.836
Cattle–goats	0.774

[TODO: Final note: report only pairs with $|\rho| \geq 0.75$ from the archived 2024 predictor screen.]

Table 8: [TODO: Final caption: pairwise overlap of outside-core high-current cells among successive snapshots and endpoints.]

Earlier year	Later year	Earlier cells	Later cells	Intersection	Union	Jaccard
2012	2016	209,148	209,150	200,069	218,229	0.917
2016	2020	209,150	209,147	201,179	217,118	0.927
2020	2024	209,147	209,146	199,881	218,412	0.915
2012	2024	209,148	209,146	199,511	218,783	0.912

[TODO: Final note: cells are the outside-core top decile of positive modeled current in each snapshot.]

Table 9: [TODO: Final caption: conditional 2030 scenario changes in selected-link least-cost path cost relative to 2024.]

Scenario	Links	Median change (%)	Minimum (%)	Maximum (%)
Low growth	45	0.00	0.00	0.00
Continuation	45	−3.36	−8.53	+6.06
High development	45	−6.94	−17.12	+8.65

[TODO: Final note: change is relative to the exact 2024 low-growth baseline; these are conditional development scenarios, not forecasts.]

Table 10: [TODO: Final caption: sensitivity of 2012–2024 selected-link cost change to vegetation weighting.]

Vegetation weighting	Links	Median absolute change (%)
Low	45	3.57
Balanced	45	6.27
High	45	9.28

[TODO: Final note: 43 of 45 links retained the same 2012–2024 change direction under all three vegetation weightings.]

Table 11: [TODO: Final caption: route-location sensitivity under the corrected vegetation inputs.]

Link set	Links	Route-years	Route-years below 80%	Affected links
All selected links	45	180	49	29
Primary-priority links	32	128	34	21

[TODO: Final note: the threshold requires at least 80 percent reciprocal route overlap within 1 km; it is a sensitivity screen, not a biological cutoff.]

it from a map legend.]

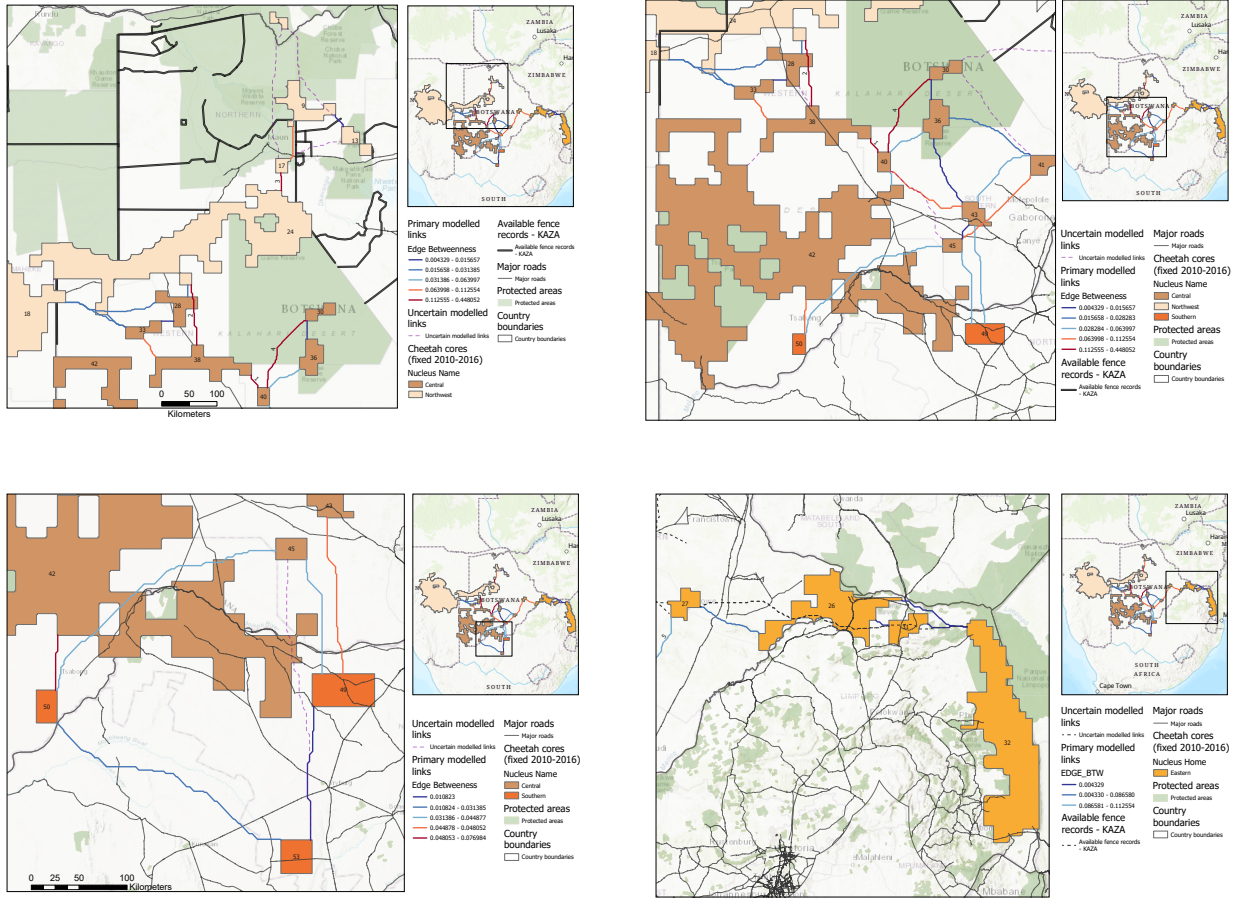


Figure 5: Regional detail maps corresponding to the modeled-link network. These panels support visual inspection of local context and are supplementary to the main network and priority-link figures.